

QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3
– Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]

Q2 Library Book Management [BL3 – Apply]

10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

CLASS Book

```
PRIVATE title : STRING  
PRIVATE author : STRING  
PRIVATE isbn : STRING  
PRIVATE isAvailable : BOOLEAN
```

```
PUBLIC METHOD setDetails(t, a, i)  
    title <- t  
    author <- a  
    isbn <- i
```

 Save Answer

Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy

Your Answer

```
author <- a
isbn <- i
isAvailable <- TRUE
END METHOD

PUBLIC METHOD issueBook( )
  IF isAvailable = TRUE
THEN
    isAvailable <- FALSE
    OUTPUT "Book Issued: ", title
  ELSE
    OUTPUT "Book Not Available: ", title
  END IF
END METHOD

PUBLIC METHOD returnBook( )
  IF isAvailable = FALSE
THEN
    isAvailable <- TRUE
    OUTPUT "Book Returned: ", title
```

 Save Answer

Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy

Your Answer

```
OUTPUT "Book Returned: ", title
ELSE
    OUTPUT "Book Already Available: ", title
END IF
END METHOD

PUBLIC METHOD getTitle()
    RETURN title
END METHOD

PUBLIC METHOD getAuthor()
    RETURN author
END METHOD

PUBLIC METHOD getISBN()
    RETURN isbn
END METHOD

PUBLIC METHOD getAvailability()
    RETURN isAvailable
END METHOD

PUBLIC METHOD displayDetails()
    OUTPUT "Title: ", getTitle()
```


Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

```

PUBLIC METHOD displayDetails()
    OUTPUT "Title: ", getTitle()
    OUTPUT "Author: ", getAuthor()
    OUTPUT "ISBN: ", getISBN()
    OUTPUT "Available: ", getAvailability()
END METHOD
    
```

END CLASS

BEGIN

DECLARE b1, b2, b3 AS Book

CALL b1.setDetails("Java Basics", "Author A ", "111")
CALL b2.setDetails("OOP Concepts", "Author B", "222")
CALL b3.setDetails("Data Structures", "Author C", "333")

CALL b1.issueBook()
CALL b1.returnBook()

CALL b2.issueBook()
CALL b2.issueBook()

CALL b3.displayDetails()

END

 Save Answer